ECON 609: Particle Filter

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Lecture # 7

From Linear to Nonlinear Models

- While most dynamic models are inherently nonlinear, the nonlinearities are often small and decision rules are approximately linear.
- One can add certain features that generate more pronounced nonlinearities.
  - stochastic volatility
  - markov switching coefficients
  - asymmetric adjustment costs
  - occasionally binding constraints

From Linear to Nonlinear Models

- Linear dynamic model leads to

$$y_t = \Psi_0(\theta) + \Psi_1(\theta)s_t + u_t, \quad u_t \sim N(0, \Sigma_u)$$
$$s_t = \Phi_1(\theta)s_{t-1} + \Phi_\epsilon(\theta)\epsilon_t, \quad \epsilon_t \sim N(0, \Sigma_\epsilon)$$

- Nonlinear dynamic model leads to

$$y_t = \Psi(s_t; \theta) + u_t, \quad u_t \sim F_u(\cdot; \theta)$$
$$s_t = \Phi(s_{t-1}, \epsilon_t; \theta), \quad \epsilon_t \sim F_\epsilon(\cdot; \theta)$$

Some Comments

- Recall: Random Walk Metropolis-Hastings
- We can conduct Bayesian estimation using the posterior.
- We need to evaluate the likelihood given a parameter θ.
- For linear models: Kalman filter
- For nonlinear models: particle filter

Particle Filter

- There are many particle filters (PF)...
- We will focus on three types:
  (1) Bootstrap PF
  (2) Generic PF
  (3) Conditionally-optimal PF

Filtering - General Idea

- State-space representation of nonlinear dynamic model
  - Measurement equation: $y_t = \Psi(s_t;\theta) + u_t, \quad u_t \sim F_u(\cdot;\theta)$
  - State transition equation: $s_t = \Phi(s_{t-1}, \epsilon_t;\theta), \quad \epsilon_t \sim F_\epsilon(\cdot;\theta)$

- Likelihood function: $p(Y_{1:T}|\theta) = \prod_{t=1}^{T} p(y_t|Y_{1:t-1},\theta)$

- Iterations:
  - Initialization at time $t-1$: $p(s_{t-1}|Y_{1:t-1},\theta)$
  - Forecasting t given $t-1$:
    1. Transition equation:
       $p(s_t|Y_{1:t-1},\theta) = \int p(s_t|s_{t-1}, Y_{1:t-1},\theta)p(s_{t-1}|Y_{1:t-1},\theta)ds_{t-1}$
    2. Measurement equation:
       $p(y_t|Y_{1:t-1},\theta) = \int p(y_t|s_t, Y_{1:t-1},\theta)p(s_t|Y_{1:t-1},\theta)ds_t$
  - Updating with Bayes theorem. Once y_t becomes available:

$$p(s_t|Y_{1:t},\theta) = p(s_t|y_t, Y_{1:t-1},\theta) = \frac{p(y_t|s_t, Y_{1:t-1},\theta)p(s_t|Y_{1:t-1},\theta)}{p(y_t|Y_{1:t-1},\theta)}$$

Bootstrap Particle Filter

(1) Initialization: Draw the initial particles from the distribution $s_0^i \sim_{i.i.d} p(s_0)$ and set $W_0^j = 1, j = 1, \ldots, M$.

(2) Recursion. For $t = 1, \ldots, T$:

- Forecasting s_t: Propagate the period $t-1$ particles $\{s_{t-1}^j, W_{t-1}^j\}$ by iterating the state-transition equation forward:
$$\tilde{s}_t^j = \Phi(s_{t-1}^j, \epsilon_t^j; \theta), \quad \epsilon_t^j \sim F_\epsilon(\cdot; \theta)$$
An approximation of $\mathbb{E}[h(s_t)|Y_{1:t-1}, \theta]$ is given by
$$\widehat{h}_{t,M} = \frac{1}{M} \sum_{j=1}^{M} h(\tilde{s}_t^j) W_{t-1}^j$$

Bootstrap Particle Filter

(1) Initialization.

(2) Recursion. For $t = 1, \ldots, T$:

- Forecasting s_t.
- Forecasting y_t. Define incremental weights $\widetilde{w}_t^j = p(y_t | \widetilde{s}_t^j, \theta)$.

  The predictive density $p(y_t | Y_{1,t-1}, \theta)$ can be approximated by

$$\widehat{p}(y_t | Y_{1:t-1}, \theta) = \frac{1}{M} \sum_{j=1}^{M} \widetilde{w}_t^j W_{t-1}^j$$

If the measurement errors $\sim N(0, \Sigma_u)$, then the incremental weights take the form

$$\widetilde{w}_t^j = (2\pi)^{-n/2} |\Sigma_u|^{-1/2} \exp\left\{ -\frac{1}{2} (y_t - \Psi(\widetilde{s}_t^j; \theta))' \Sigma_u^{-1} (y_t - \Psi(\widetilde{s}_t^j; \theta)) \right\}$$

where n denotes the dimension of y_t.

Bootstrap Particle Filter

(1) Initialization.

(2) Recursion. For $t = 1, \ldots, T$:

- Forecasting s_t.
- Forecasting y_t. Define incremental weights $\tilde{w}_t^j = p(y_t|\tilde{s}_t^j, \theta)$.
- Updating. Define normalized weights

$$\widetilde{W}_t^j = \frac{\tilde{w}_t^j W_{t-1}^j}{\frac{1}{M}\sum_{j=1}^M \tilde{w}_t^j W_{t-1}^j}$$

The larger the variance of the particle weights, the more inefficient the approximation. Define the effective sample size

$$\widehat{ESS}_n \equiv M / \left(\frac{1}{M} \sum_{j=1}^{M} (\widetilde{W}_i^j)^2 \right)$$

Bootstrap Particle Filter

(1) Initialization.

(2) Recursion. For $t = 1, \ldots, T$:

- Forecasting s_t.
- Forecasting y_t.
- Updating.
- Selection. If the effective sample size (ESS) falls below the half of the number of particles M, we do resampling.

  Let $\{s_t^j\}_{j=1}^M$ denote M i.i.d. draws from a multinomial distribution characterized by support points and weights $\{\tilde{s}_t^j, \widetilde{W}_t^j\}$ and set $W_t^j = 1$ for $j = 1, \ldots, M$.

Likelihood Approximation

- The approximation of the log likelihood function is given by

$$ln\widehat{p}(Y_{1:T}|\theta) = \sum_{t=1}^{T} ln\left(\frac{1}{M}\sum_{j=1}^{M}\widetilde{w}_t^j W_{t-1}^j\right)$$

- Bootstrap filter needs non-degenerate $p(y_t|s_t,\theta)$ for incremental weights to be well-defined.